

AYBUStore: Intelligent Campus E-Commerce Ecosystem

Supplementary Technical Report for TÜBİTAK 1001 Proposal

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Abstract

This report details the technical architecture, methodology, and requirement specifications for AYBUStore, a smart e-commerce platform developed for Ankara Yıldırım Beyazıt University. Utilizing a modern stack of Next.js 14 and Firebase, the project integrates AI-driven personalization and real-time inventory management to solve physical campus commerce inefficiencies.

1 Introduction & Problem Definition

The AYBUStore project addresses the lack of a centralized digital ecosystem for university-branded merchandise and department-specific materials. Existing solutions fail to provide real-time stock transparency, personalized apparel preview, or AI-integrated shopping assistance.

2 System Architecture

We adopt a **Component-Based Architecture** using Next.js App Router for optimal performance and SEO. The backend is powered by Firebase, providing a serverless real-time database and secure authentication.

2.1 Use Case Analysis

The following diagram illustrates the interactions between students, staff, and administrators within the ecosystem.

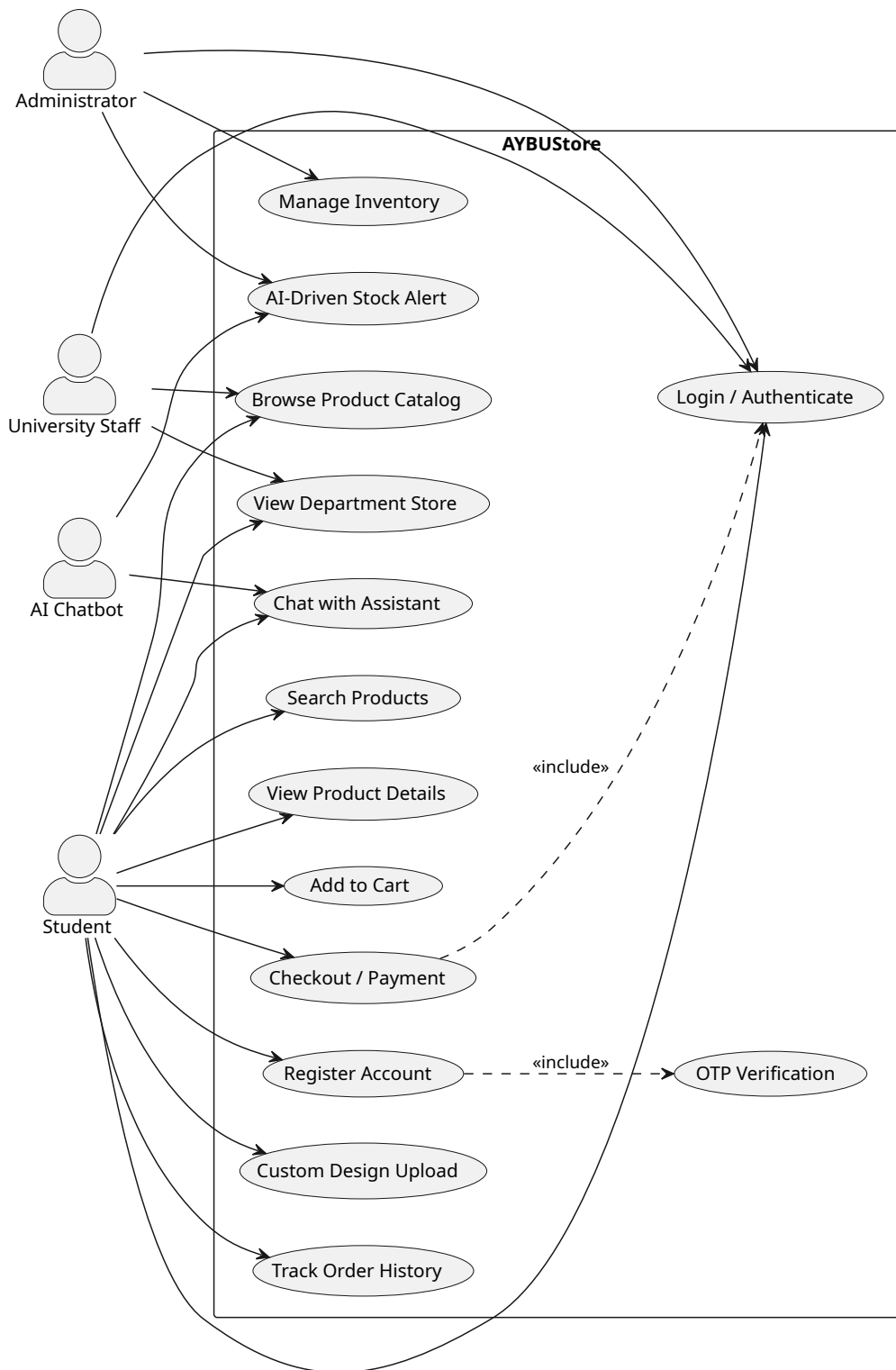


Figure 1: AYBUStore Use Case Diagram (Vector)

2.2 Domain Modeling

The system's data structure is designed with modularity in mind, ensuring scalability as new faculties are added.

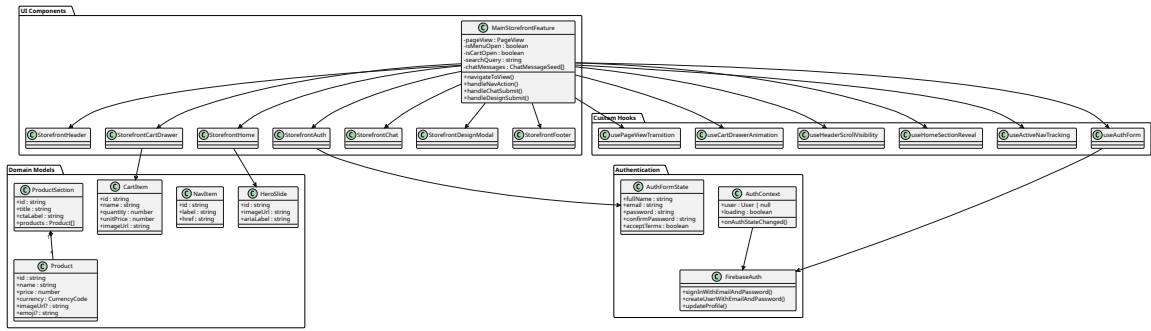


Figure 2: System Domain Class Diagram

3 Core Processes

3.1 Authentication & OTP Flow

Security is managed via a multi-stage authentication process, verified by a 6-digit one-time password (OTP) sent to university emails.

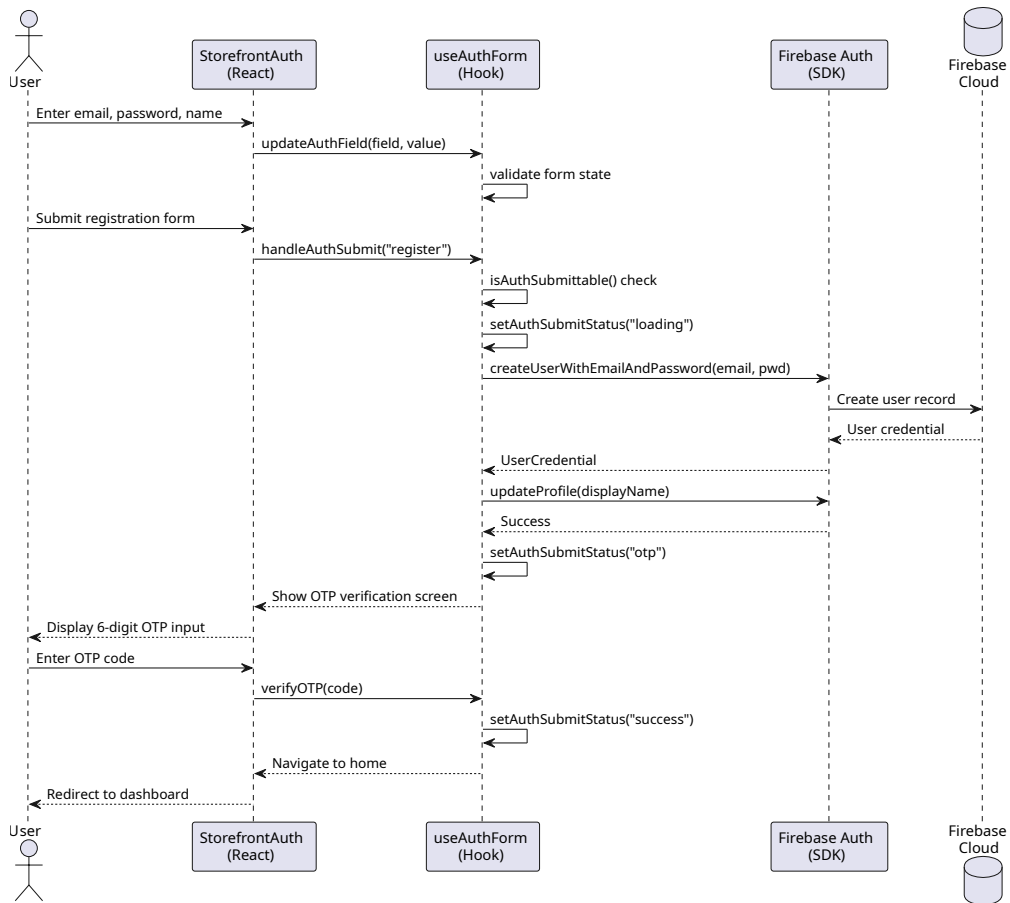


Figure 3: Registration and Security Verification Sequence

3.2 AI Personalization

The AI Shopping Assistant utilizes LLM-based intent analysis to provide context-aware product recommendations.

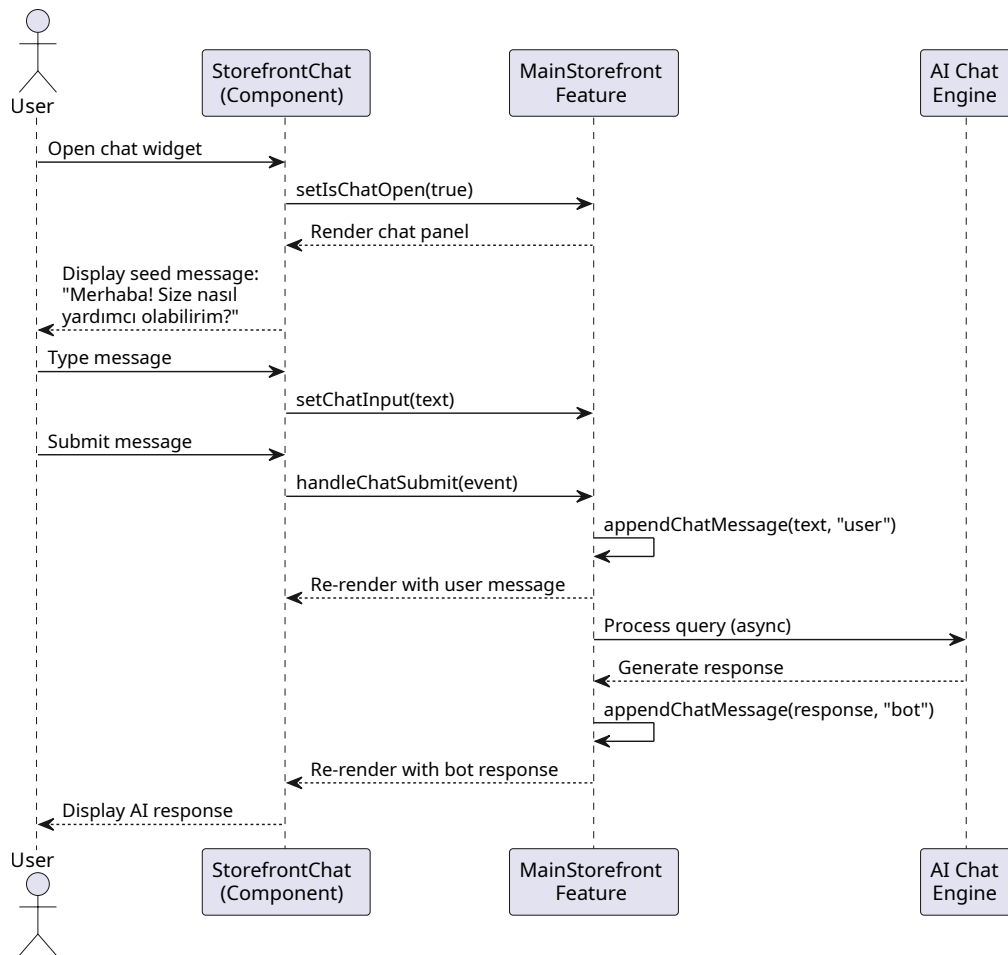


Figure 4: AI-Driven Assistant Interaction Flow

4 Operational Design

4.1 Shopping Lifecycle

The activity flow below describes the transition from browsing to custom design and final checkout.

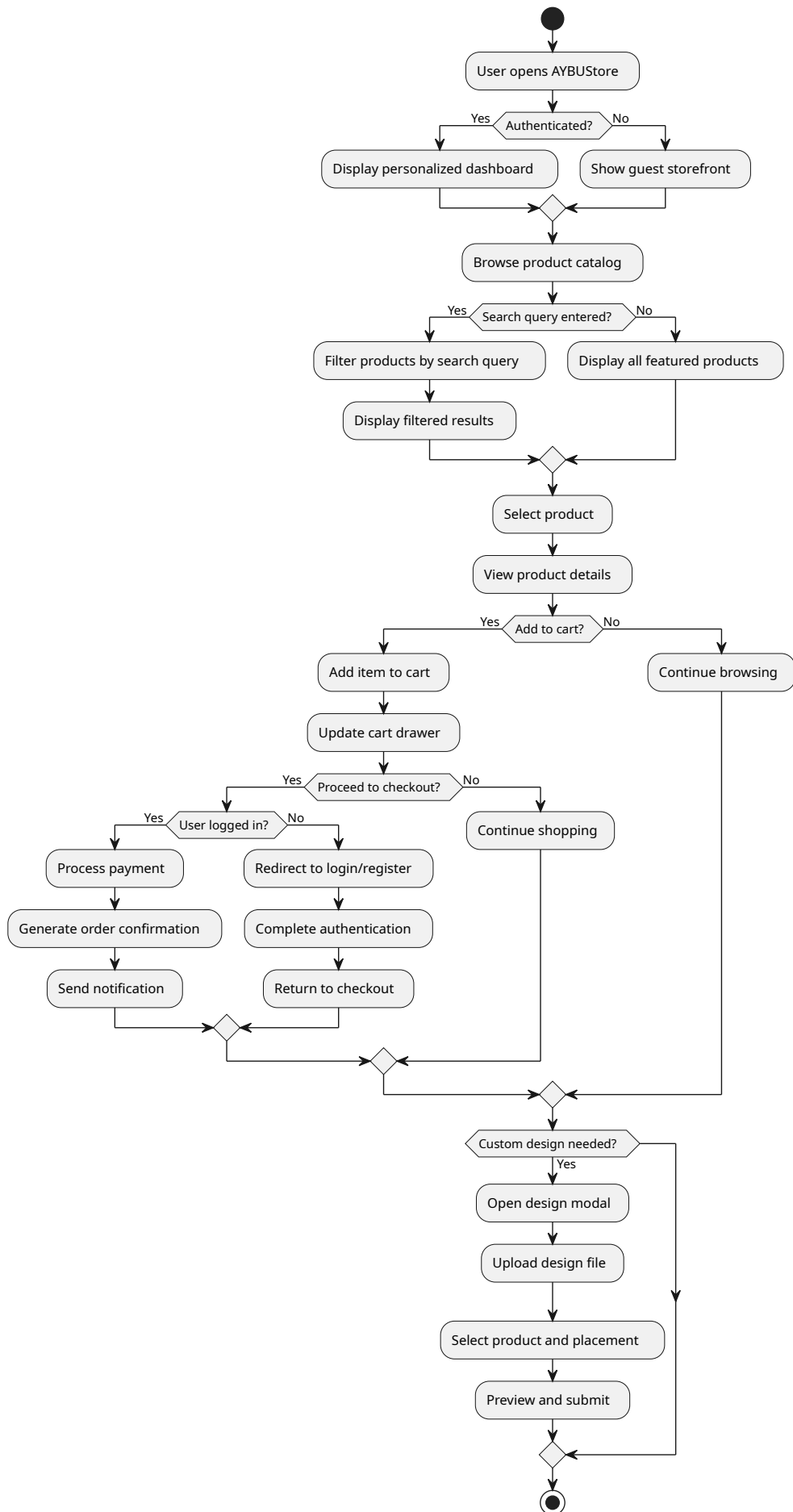


Figure 5: User Activity Lifecycle Diagram

5 Implementation Plan

The project follows an **Agile/Scrum** framework divided into four primary sprints:

1. **Sprint 1:** Core Infrastructure & Design System.
2. **Sprint 2:** Auth, Product Catalog & Cart Engine.
3. **Sprint 3:** AI Integration & Custom Design Module.
4. **Sprint 4:** Performance Optimization & User Acceptance Testing.

6 Conclusion

AYBUStore provides a scalable and intelligent solution for campus commerce, leveraging modern web technologies to enhance the university's digital brand.